PINGCHENG JIAN

Post-Doc research associate, Department of Computing, Imperial College London
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APPOINTMENTS

Imperial College London *Nov. 2025 - present*
Postdoc in the Department of Computing Host: Edward Johns

EDUCATION

Duke University *Sep. 2021 - Sep. 2025*
Ph.D. in Electrical and Computer Engineering Advisors: Michael Zavlanos, Boyuan Chen

Tsinghua University *Aug. 2016 - Jul. 2021*
B.E. in Automation Advisors: Huaping Liu, Mingguo Zhao
Transfer major from Mechanical Engineering (2016-2018) to Automation (2018-2021)

RESEARCH INTERESTS

My area of research is robot learning. I am currently working on the ARIA-funded project 'Democratising Co-Design of Hardware and Control for Robot Dexterity' project in the Robot Dexterity program. My research focuses on developing robot learning methods for complex behavior, efficient generalization, and strong robustness.

PUBLICATIONS

($*$ equal contributions, $**$ alphabetical order. Up-to-date list can be found in my Google Scholar page.)

Journal Papers

1. **[TMLR'25]** Pingcheng Jian, Xiao Wei, Yanbaihui Liu, Samuel A. Moore, Michael M. Zavlanos, Boyuan Chen, "LAPP: Large Language Model Feedback for Preference-Driven Reinforcement Learning", *Transactions on Machine Learning Research (TMLR)*, 2025. [video] [arxiv]
2. **[TMLR'24]** Pingcheng Jian, Easop Lee, Zachary I. Bell, Michael M. Zavlanos, Boyuan Chen, "Perception Stitching: Zero-Shot Perception Encoder Transfer for Visuomotor Robot Policies", *Transactions on Machine Learning Research (TMLR)*, 2024. [website] [video] [arxiv]

Conference Papers

3. **[CoRL'23]** Pingcheng Jian, Easop Lee, Zachary Bell, Michael M. Zavlanos, Boyuan Chen, "Policy Stitching: Learning Transferable Robot Policies", *Conference on Robot Learning (CoRL)*, 2023. [website] [CoRL talk] [video] [arXiv]
4. **[ICRA'21]** Pingcheng Jian*, Chao Yang*, Di Guo, Huaping Liu, Fuchun Sun, "Adversairial Skil Learning for Robust Manipulation", *International Conference on Robotics and Automation (ICRA)*, 2021. [pdf] [video]
5. **[L4DC'23]** Reza Khodayi-mehr, Pingcheng Jian, Michael M. Zavlanos, "Physics-Guided Active Learning of Environmental Flow Fields", *Learning for Dynamics and Control (L4DC)*, 2023. [pdf]

Workshop Papers

6. **[ICML-W'21]** Minghao Zhang*, Pingcheng Jian*, Yi Wu, Huazhe Xu, Xiaolong Wang, "DAIR: Disentangled Attention Intrinsic Regularization for Safe and Efficient Bimanual Manipulation", *International Conference on Machine Learning (ICML) Workshop: Reinforcement Learning for Real Life*, 2021. [website]

PATENT

US Patent (pending): Large Language Model-Assisted Preference Prediction for Reinforcement Learning (Application No. 63/751,383, filed on January 30, 2025; Attorney Docket No.: DU8724PROV)

Chinese Invention Patent: A Method and System for Generating Packing Solutions for Items in a Logistics Warehouse (Publication No.: CN110443549B, Application No.: 2019106808700)

PROJECTS

Quadruped Spider Robot, course project of *CS590 - Robot Studio* at Duke University
Project Video: [video]
I built all the hardware and software of this spider robot from scratch. I drew the parts of this robot with SOLIDWORKS and then printed them with 3D printer. The controller of this robot is a Raspberry Pi, and the control algorithm is written in Python. Then I generated the .urdf files of this robot and built the simulator of this robot with PyBullet.

ACADEMIC SERVICE

Reviewer:
Conference on Robot Learning (CoRL), 2024
Conference on Robot Learning (CoRL), 2025

TEACHING EXPERIENCES

Teaching Assistant at Duke University
ECE 382L/ME 344L - Control of Dynamic Systems [lecture] [recitation]
ECE 689/COMPSCI 676 - Advanced Topics in Deep Learning

ADVISING AND MENTORSHIP

Easop Lee, Duke Undergraduate Student in ECE (2022). Next: Ph.D. at Duke ECE
Xiao Wei, Duke CS Master (2024 - present)
Yongjae Lee, Duke ME Master (2024 - present)

AWARDS

Second Prize, 2015 National High School Mathematics Joint Competition, China
Second Prize, Guangdong Regional Selection of the 2015 National High School Mathematics Joint Competition, China
Second Prize, 2014 High School Mathematical Contest in Modeling (HiMCM), USA

SKILLS

Programming Languages
Python, Matlab, C, C++, C#, HTML, Assembly Language
Machine Learning
Pytorch, Reinforcement Learning, Supervised Learning
Robotic Software
PyBullet, MuJoCo, Isaac Gym
Robotic Hardware
SOLIDWORKS, 3D printing